

- 1 **B** Diameter of a ping-pong is 40 mm, so radius is 2 cm.

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(0.02)^3 = 3.35 \times 10^{-5} \text{ m}^3$$

$$U = V\rho g = 3.35 \times 10^{-5} \times 1000 \times 9.81 = 0.33 \text{ N}$$

- A** Let's take a small 1 tonne (10 000 kg) bus moving at 70 km h⁻¹ (19.44 m s⁻¹).

$$\text{KE} \approx \frac{1}{2}(10000)(19)^2 = 1.8 \text{ MJ}$$

- C** A domestic bulb is around 25 to 100 W.

D Energy of a typical microwave photon = hc / λ
 $= (6.63 \times 10^{-34})(3.00 \times 10^8) / (3 \times 10^{-2}) = 6.63 \times 10^{-24} \text{ J}.$

- 2 **B** Shortest path is perpendicularly across. Hence resultant velocity should be in this direction. To